

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and debugged embedded software for **mixed-signal ATE platforms**, supporting **silicon functionality** and characterization for new product releases.
- Designed and executed test solutions on **high-performance ATE**, achieving **95% test coverage** for **analog and digital electronics** in ultra-high-volume production.
- Automated test data analysis and yield reporting using **Python scripting**, reducing manual analysis time by **40%** and improving defect identification speed.
- Collaborated with cross-functional NP development and test teams to define **DFT (Design for Testability)** strategies, enhancing product quality and reducing manufacturing costs.
- Managed change control processes for test solution updates, coordinating **User Acceptance Testing (UAT)** and providing post-go-live support for offshore production releases.
- Implemented **containerization using Docker** for test environment setup, enabling consistent and repeatable deployments across development and production.

SKILLS

Programming Languages – Python, Java, C, Visual Basic.

Hardware & Embedded – Analogue Electronics, Digital Electronics, Mixed-Signal ATE, DFT, Silicon Functionality, Characterization.

Software Development Tools – Docker, Git, Jenkins, Jira, Confluence, Linux.

Data Analysis & Automation – Python Scripting, Data Pipelines, ETL, Scheduled Reporting, Yield Analysis.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch).

Testing Methodologies – Test Development, Multi-site Probe & Test Solutions, Production Release, Debug, UAT Support.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Automated Test Solution for Capacitive Sensors

- Developed automated test sequences in **Python** for validating capacitive sensing ICs on a **mixed-signal ATE platform**.
- Integrated **Java-based drivers** to interface with test hardware, enabling real-time data acquisition and analysis of sensor performance.

- Designed test procedures to ensure compliance with **product specifications**, contributing to the early detection of design flaws before tape-out.

Production Yield Analysis Dashboard

- Created a **Power BI dashboard** to visualize yield data and key performance indicators from high-volume production test runs.
- Developed **DAX measures** and Power Query transformations for ETL, consolidating data from multiple test sources and enabling detailed trend analysis.
- Collaborated with operations teams to refine reporting requirements, improving the visibility of manufacturing bottlenecks by **30%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)